

# Computer Networks



## Overview of the Internet Protocol layering (Book 1 Ch 1)

### Application Layer

- Application-Layer Paradigms
- Client-server applications
  - World Wide Web and HTTP
  - FTP
  - Electronic Mail
  - DNS
- Peer-to-peer paradigm
  - P2P Networks
- Case study: BitTorrent (Book 1 Ch 2)

## ① Introduction

- Providing Services
- Application-Layer Paradigms

## ② Standard Client-Server Applications

- World Wide Web and HTTP
- FTP
- Electronic Mail
- Domain Name System (DNS)

## ③ Peer-to-Peer Paradigm

- P2P Networks

- Original Web designed as:
  - Stateless entity
- Stateless communication:
  - Client sends request
  - Server sends response
  - Connection information forgotten
- Original Web purpose:
  - Retrieve public documents
- Modern Web applications require:
  - Client information retention

# Need for Cookies

- Cookies introduced to:
  - Remember client information
- Required by applications such as:
  - Electronic shopping
  - Restricted-access websites
  - Web portals
  - Advertising agencies
- Cookies provide:
  - State management in Web communication

# Electronic Shopping and Cookies

- E-commerce websites allow users to:

- Browse products
- Select items
- Store items in shopping cart
- Pay online

- Cookies store:

- Shopping cart information

- Cookie updated whenever:

- New item selected

# Restricted Access and Portals

- Some websites allow access only to:
  - Registered users
- Cookies identify:
  - Authorized clients
- Web portals use cookies to:
  - Store user preferences
  - Remember favorite pages
- Returning users receive:
  - Customized web pages

# Advertising and Cookies

- Advertising agencies use cookies to:
  - Track user behavior
- Banner advertisements may contain:
  - URL of advertising agency
- Clicking advertisement sends:
  - Request to advertising server
- Cookies help agencies:
  - Build user interest profiles

# Creating and Storing Cookies

- ❶ Server receives client request
  - ❷ Server creates cookie containing client information
  - ❸ Cookie included in server response
  - ❹ Browser stores cookie in cookie directory
- Cookie directory organized by:
    - Server domain name

# Contents of a Cookie

- Cookie may contain:
  - Client domain name
  - User information
  - Registration number
  - Timestamp
  - Other implementation details
- Cookie created by:
  - Server

# Using Cookies

- Browser checks:
  - Cookie directory before sending request
- If matching cookie found:
  - Cookie attached to request
- Server recognizes:
  - Returning client
- Cookie contents:
  - Not disclosed to user

# Cookies in E-Commerce

- Cookie stores:
  - Item number
  - Unit price
- Cookie updated after:
  - Each item selection
- During checkout:
  - Cookie retrieved
  - Total charge calculated

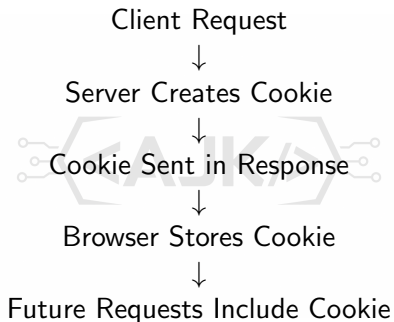
# Cookies for Registered Users

- Website sends cookie:
  - During first registration
- Future access allowed only for:
  - Clients sending valid cookie
- Cookie acts as:
  - User identification mechanism

# Cookies and Privacy Issues

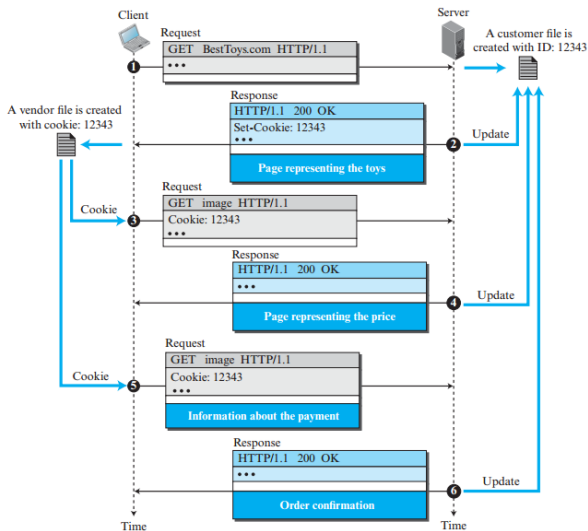
- Advertising agencies may:
  - Track browsing behavior
- User interests stored in:
  - Databases
- Information may be:
  - Sold to third parties
- Cookie tracking creates:
  - Privacy concerns
- Need for:
  - Privacy regulations

# Cookie Operation Illustration



# Example 2.9

Figure 2.15 Example 2.9



## Example 2.9: Cookies in Electronic Shopping

- Example demonstrates:
  - Use of cookies in e-commerce
- Electronic store:
  - BestToys
- Shopper accesses:
  - BestToys website
- Cookies help server:
  - Remember customer information



# Initial Client Request

- Shopper browser sends:
  - Request to BestToys server
- Server creates:
  - Empty shopping cart
- Shopping cart assigned:
  - Unique ID
- Example cart ID:
  - 12343

# Server Response with Cookie

- Server response contains:
  - Images of toys
  - Links for toy selection
- Response also includes:
  - Set-Cookie header
- Cookie value:
  - 12343
- Browser stores cookie:
  - In file named BestToys

# Cookie Privacy

- Cookie stored by:
  - Browser
- Cookie not revealed to:
  - Shopper
- Browser automatically handles:
  - Cookie storage
  - Cookie transmission

# Selecting a Toy

- Shopper clicks:
  - Selected toy
- Browser sends:
  - New request
- Request contains:
  - Cookie header line
- Cookie value included:
  - 12343



# Server Identifies Returning Client

- Server reads:
  - Cookie value 12343
- Server understands:
  - Shopper is returning client
- Server searches for:
  - Shopping cart with ID 12343
- Selected toy added to:
  - Existing shopping cart

# Payment Processing

- Server sends response containing:
  - Total price
  - Payment request
- Shopper provides:
  - Credit card information
- Client sends another request:
  - Including cookie value 12343

# Order Confirmation

- Server receives:
  - Request with cookie 12343
- Server verifies:
  - Shopping cart information
- Server accepts:
  - Order
  - Payment
- Confirmation sent back:
  - In response message

# Future Visits to the Store

- Browser sends cookie again:
  - During future visits
- Server retrieves:
  - Stored customer information
- Store remembers:
  - Shopper details
  - Previous interactions
- Cookies provide:
  - Personalized shopping experience

## Cookie Workflow in Example 2.9



# Advantages of Cookies in E-Commerce

- Cookies help:
  - Maintain shopping sessions
  - Track customer selections
  - Store user preferences
- Improves:
  - User convenience
  - Shopping efficiency
- Eliminates need for:
  - Re-entering information repeatedly

# Web Caching and Proxy Server

- HTTP supports:
  - Proxy servers
- Proxy server:
  - Computer that stores copies of recent responses
- Purpose of caching:
  - Reduce repeated requests to original server
- Cached responses reused for:
  - Future client requests

# Working of a Proxy Server

- Client sends request to:
  - Proxy server
- Proxy server checks:
  - Local cache memory
- If response found:
  - Response sent directly to client
- If response not found:
  - Proxy forwards request to original server
- Incoming response:
  - Stored in proxy cache
  - Sent to client

# Advantages of Proxy Servers

- Reduces:
  - Load on original server
- Decreases:
  - Network traffic
- Improves:
  - Latency
  - Response time
- Frequently requested pages:
  - Delivered faster



# Client Configuration for Proxy Server

- Client must be configured to:
  - Access proxy server
- Client does not contact:
  - Target server directly
- Proxy server acts as:
  - Intermediate system

# Dual Role of Proxy Server

- Proxy server acts as:
  - Server
  - Client
- When cached response exists:
  - Acts as server
  - Sends response to client
- When cached response absent:
  - Acts as client
  - Sends request to target server

# Proxy Server Communication Flow

- ① Client sends request to proxy server
- ② Proxy checks cache
- ③ If cached:
  - Response sent to client
- ④ If not cached:
  - Proxy sends request to target server
- ⑤ Target server sends response
- ⑥ Proxy stores response in cache
- ⑦ Proxy forwards response to client

# Proxy Server Location

- Proxy servers usually located:
  - Near client side
- Multiple proxy levels possible:
  - Client level
  - Company level
  - ISP level
- Creates:
  - Hierarchy of proxy servers



# Client-Level Proxy Server

- Single client computer can act as:
  - Small proxy server
- Stores:
  - Frequently accessed responses
- Benefits:
  - Faster local access

# Company-Level Proxy Server

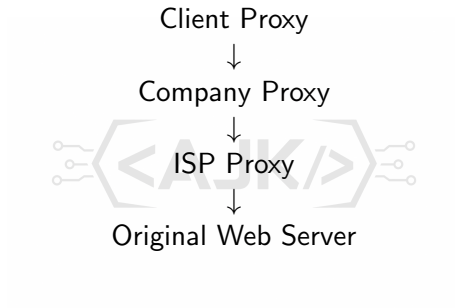
- Proxy installed on:
  - Company LAN
- Purpose:
  - Reduce incoming traffic
  - Reduce outgoing traffic
- Shared cache improves:
  - Organization-wide efficiency

# ISP-Level Proxy Server

- ISP may install:
  - Large proxy server
- Used for:
  - Many customers
- Benefits:
  - Reduced ISP traffic
  - Faster content delivery



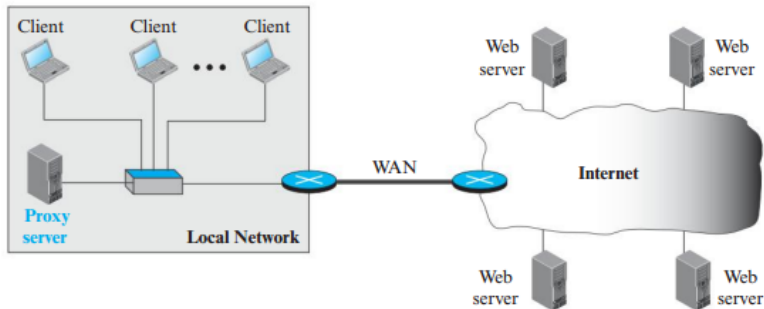
# Hierarchy of Proxy Servers



- Multiple proxy levels improve caching efficiency

## Example 2.10

**Figure 2.16** *Example of a proxy server*



## Example 2.10: Proxy Server in Local Network

- Example demonstrates:
  - Use of proxy server in local network
- Local network examples:
  - Campus network
  - Company network
- Proxy server installed:
  - Inside local network

# Client Request Handling

- HTTP request created by:
  - Client browser
- Request first directed to:
  - Proxy server
- Client does not contact:
  - Web server directly
- Proxy acts as:
  - Intermediate system

# When Web Page Exists in Cache

- Proxy server checks:
  - Cache memory
- If requested web page exists:
  - Proxy sends response directly
- No need to contact:
  - Internet web server
- Result:
  - Faster response
  - Reduced traffic

# When Web Page Does Not Exist in Cache

- If page not found in cache:
  - Proxy acts as client
- Proxy sends request to:
  - Web server on Internet
- Web server processes:
  - HTTP request
- Response returned to:
  - Proxy server

# Caching the Response

- Proxy server receives:
  - Response from Internet server
- Proxy creates:
  - Copy of response
- Copy stored in:
  - Proxy cache
- Response then forwarded to:
  - Requesting client



# Benefits of Proxy Caching

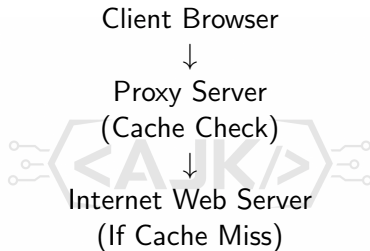
- Reduces:
  - Internet bandwidth usage
  - Traffic load
- Improves:
  - Access speed
  - Response time
- Frequently requested pages:
  - Retrieved locally
- Decreases load on:
  - Original web server



# Proxy Server Communication Flow

- ① Client browser sends HTTP request
- ② Proxy server checks cache
- ③ If cached:
  - Send page to client
- ④ If not cached:
  - Proxy requests page from Internet server
- ⑤ Web server sends response
- ⑥ Proxy stores copy in cache
- ⑦ Proxy forwards response to client

## Illustration of Example 2.10



- Proxy server acts between clients and Internet

# Key Points from Example 2.10

- Proxy server installed in:
  - Local network
- Proxy stores:
  - Copies of requested web pages
- Future requests served from:
  - Local cache
- Results:
  - Faster access
  - Reduced Internet traffic
  - Lower server load

# Cache Update

- Important issue in proxy caching:
  - How long cached response should remain stored
- Cached data eventually:
  - Deleted or replaced
- Different cache update strategies:
  - Based on type of website
  - Based on modification time
- Example:
  - News websites update content daily
  - Proxy may cache news page for one day
- HTTP headers may contain:
  - Last modification time
- Proxy server uses header information to:
  - Estimate validity period of cached data

# HTTP Security and HTTPS

- HTTP itself:
  - Does not provide security
- Security achieved using:
  - Secure Socket Layer (SSL)
- HTTP running over SSL called:
  - HTTPS
- HTTPS provides:
  - Confidentiality
  - Client authentication
  - Server authentication
  - Data integrity
- HTTPS used for:
  - Secure web communication
  - Online banking
  - E-commerce transactions

